

FINANCIAL FRAUD DETECTION USING R

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INTRODUCTION

Financial fraud has become a serious issue in modern banking systems and online transactions. Fraudulent activities can lead to financial losses, security threats, and reduced customer trust. Detecting suspicious transactions quickly is very important for banks and financial institutions.

This project focuses on the application of R programming for detecting fraudulent financial transactions. The system analyzes transaction details such as transaction amount, transaction time, and transaction location to determine whether a transaction is normal or fraudulent.

The program uses predefined logical conditions to identify suspicious activities. Transactions with unusually high amounts, transactions made during late-night hours, or transactions occurring from foreign locations are marked as fraudulent.

The project also provides graphical analysis using bar plots and pie charts to help users visualize fraud statistics more effectively. Overall, this project demonstrates how programming and data analysis techniques can be applied in the field of finance for fraud prevention and monitoring.

DATA DESCRIPTION

The data used in this project is user-generated transaction data entered during program execution. The dataset includes the following variables:

- **Transaction ID** – Unique identification number for each transaction.
- **Transaction Amount** – Amount involved in the transaction.
- **Transaction Time** – Time at which the transaction occurs (0–23 hours).
- **Transaction Location** – Location from where the transaction is made.
- **Transaction Status** – Classification of transaction as “Fraud” or “Normal”.

The variables used in this project are numerical and categorical in nature, making them suitable for analysis and classification using conditional programming techniques in R.

METHODOLOGY AND GRAPHS

1. Fraud Detection Conditions

The system applies logical conditions to identify fraudulent transactions.

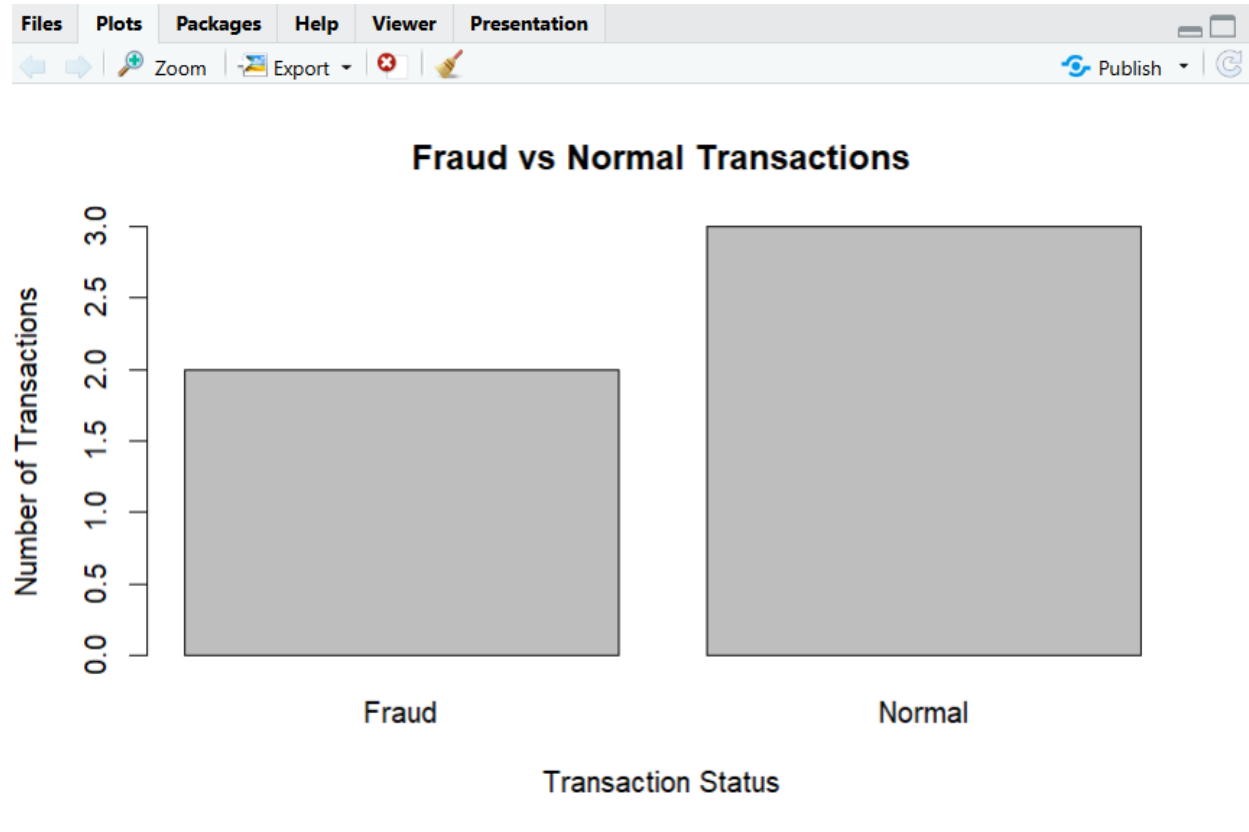
A transaction is marked as **Fraud** if:

- The transaction amount is greater than 100000.
- The transaction occurs between 12 AM and 4 AM.

- The transaction location is “Foreign”.

Otherwise, the transaction is classified as **Normal**.

Figure 1: Fraud vs Normal Transactions



The bar chart compares the number of fraudulent and normal transactions detected by the system.

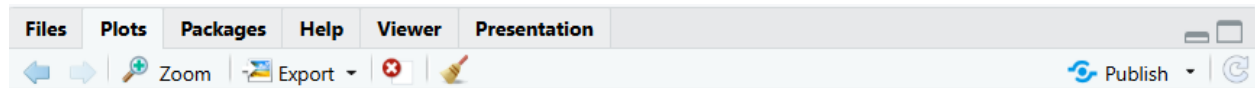
INTERPRETATION:

- Higher bars indicate a greater number of transactions in that category.
- The graph helps users quickly identify the proportion of fraud cases compared to normal cases.
- It provides a simple visual summary of transaction classification.

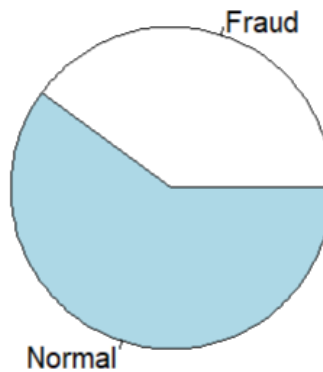
2. Fraud Percentage Analysis

The project also calculates the percentage of fraudulent transactions using total transaction records.

Figure 2: Fraud Transaction Percentage



Fraud Transaction Percentage



The pie chart visually represents the percentage distribution of fraud and normal transactions.

INTERPRETATION:

- Each section of the pie chart represents the proportion of transaction categories.
- A larger fraud section indicates a higher fraud risk.
- The graph improves understanding of overall transaction behavior.

FEATURES OF THE PROJECT

- User-friendly transaction input system.
- Automatic fraud detection mechanism.
- Fraud percentage calculation.
- Graphical representation using bar plots and pie charts.
- Simple and efficient implementation in R programming language.

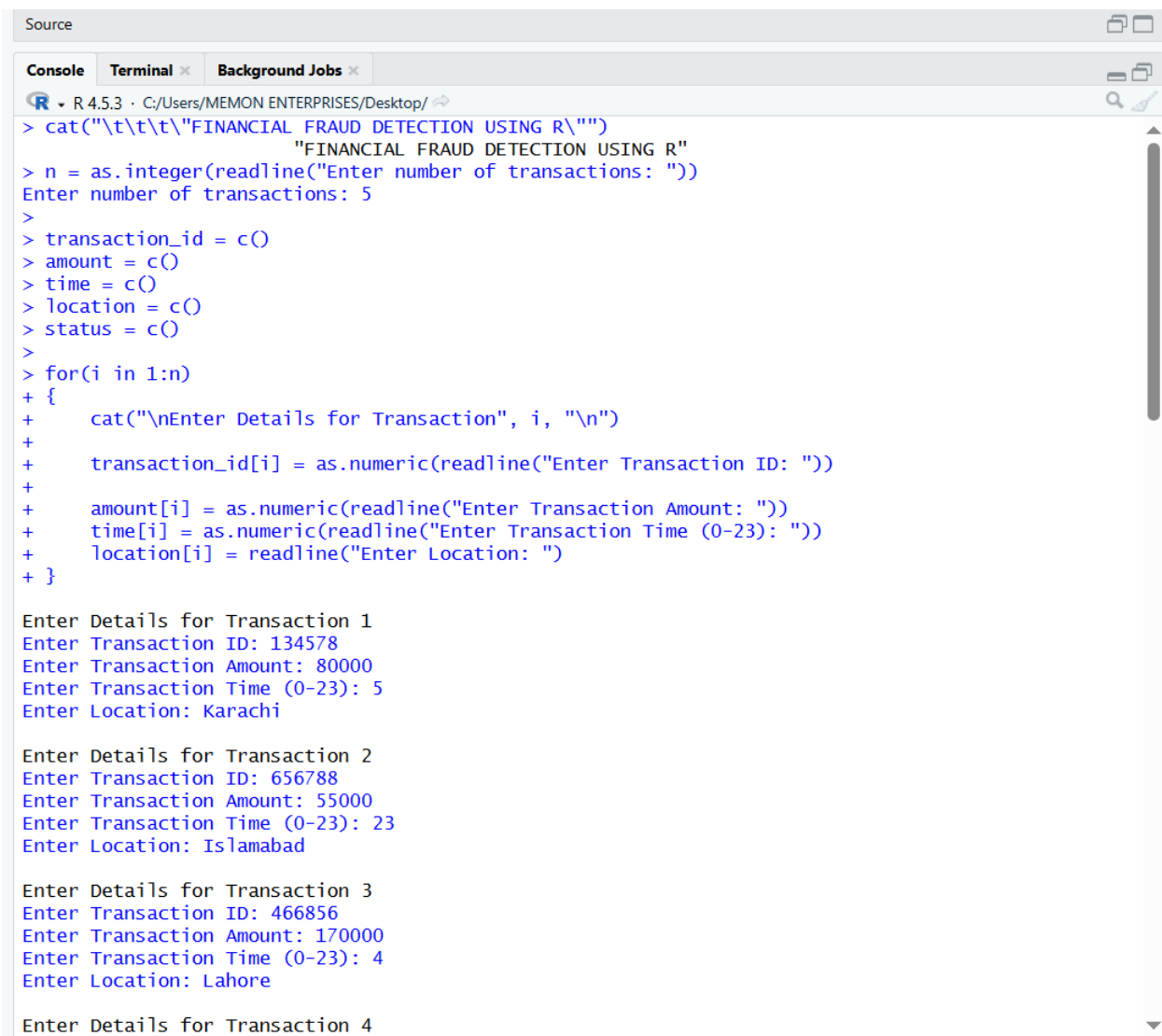
CONCLUSION

This project successfully applies R programming techniques for financial fraud detection. By analyzing transaction amount, time, and location, the system identifies suspicious transactions and classifies them as fraud or normal.

The use of graphs and statistical summaries makes the analysis more understandable and visually effective. The project highlights the importance of automated fraud detection systems in financial institutions for reducing risks and improving transaction security.

Overall, this study demonstrates how programming can support financial monitoring and fraud prevention in real-world applications.

APPENDIX



```
Source
Console Terminal Background Jobs
R 4.5.3 · C:/Users/MEMON ENTERPRISES/Desktop/
> cat("\t\t\t\t\"FINANCIAL FRAUD DETECTION USING R\"")
"FINANCIAL FRAUD DETECTION USING R"
> n = as.integer(readline("Enter number of transactions: "))
Enter number of transactions: 5
>
> transaction_id = c()
> amount = c()
> time = c()
> location = c()
> status = c()
>
> for(i in 1:n)
+ {
+   cat("\nEnter Details for Transaction", i, "\n")
+   transaction_id[i] = as.numeric(readline("Enter Transaction ID: "))
+   amount[i] = as.numeric(readline("Enter Transaction Amount: "))
+   time[i] = as.numeric(readline("Enter Transaction Time (0-23): "))
+   location[i] = readline("Enter Location: ")
+ }

Enter Details for Transaction 1
Enter Transaction ID: 134578
Enter Transaction Amount: 80000
Enter Transaction Time (0-23): 5
Enter Location: Karachi

Enter Details for Transaction 2
Enter Transaction ID: 656788
Enter Transaction Amount: 55000
Enter Transaction Time (0-23): 23
Enter Location: Islamabad

Enter Details for Transaction 3
Enter Transaction ID: 466856
Enter Transaction Amount: 170000
Enter Transaction Time (0-23): 4
Enter Location: Lahore

Enter Details for Transaction 4
```

```
Source
Console Terminal Background Jobs
R 4.5.3 · C:/Users/MEMON ENTERPRISES/Desktop/
Enter Transaction ID: 564538
Enter Transaction Amount: 60000
Enter Transaction Time (0-23): 23
Enter Location: Hyderabad

Enter Details for Transaction 5
Enter Transaction ID: 654546
Enter Transaction Amount: 39000
Enter Transaction Time (0-23): 8
Enter Location: Foreign
> for(i in 1:n){
+   if(amount[i] > 100000 |
+     (time[i] >= 0 & time[i] <= 4) |
+     location[i] == "Foreign"){
+     status[i] = "Fraud"
+   }
+   else{
+     status[i] = "Normal"
+   }
+ }
> cat("\t\t\t\"TRANSACTION REPORT\"")
"TRANSACTION REPORT"
> for(i in 1:n){
+   cat("\nTransaction ID =", transaction_id[i])
+   cat("\nAmount =", amount[i])
+   cat("\nTime =", time[i])
+   cat("\nLocation =", location[i])
+   cat("\nStatus =", status[i])
+ }

Transaction ID = 134578
Amount = 80000
Time = 5
Location = Karachi
```

```
Source
Console Terminal Background Jobs
R 4.5.3 · C:/Users/MEMON ENTERPRISES/Desktop/
Status = Normal
Transaction ID = 656788
Amount = 55000
Time = 23
Location = Islamabad
Status = Normal
Transaction ID = 466856
Amount = 170000
Time = 4
Location = Lahore
Status = Fraud
Transaction ID = 564538
Amount = 60000
Time = 23
Location = Hyderabad
Status = Normal
Transaction ID = 654546
Amount = 39000
Time = 8
Location = Foreign
Status = Fraud>
> fraud_count = sum(status == "Fraud")
>
> normal_count = sum(status == "Normal")
>
> cat("\nTotal Fraud Transactions =", fraud_count)

Total Fraud Transactions = 2>
> cat("\nTotal Normal Transactions =", normal_count)

Total Normal Transactions = 3>
> total_transactions = length(status)
>
> fraud_percentage = (fraud_count / total_transactions) * 100
>
> cat("\nFraud Percentage =", fraud_percentage,"%")

Fraud Percentage = 40 %
> barplot(table(status),
+       main = "Fraud vs Normal Transactions",
+       xlab = "Transaction Status",
+       ylab = "Number of Transactions")
>
> pie(table(status),
+     main = "Fraud Transaction Percentage")
> |
```